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# BSc (Hons) in Information Technology Specializing Data Science

## Research Project - IT4010 MLOPs Report

**Group ID**: R26-DS-010

**Project Title**: Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

1. **IT22178640**: AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution.
2. **IT22057488**: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction.
3. **IT22294098**: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs
4. **IT22289384**: Personalized Sign Language Tutoring System for Caregivers

| | | IT22178640 | IT22057488 | IT22294098 | IT22289384 |
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| **Data Pipeline** | **Data Sources:** | **TILES-2018 (Tracking Individual Performance with Sensors):** A research dataset collected at USC Keck Hospital, Los Angeles, by the Signal Analysis and Interpretation Laboratory, University of Southern California (principal investigator Dr. Shrikanth Narayanan; Mundnich et al.). It followed 212 hospital workers for about 10 weeks each.<br><br>Access was approved through a signed Data Use Agreement with the University of Southern California (signed by | ***NTU RGB+D*** – public research dataset, but access had to be requested from the dataset owners before use. I used 4 action classes and retained 3,777 processed sequences after filtering.<br><br>***UR Fall Detection*** – publicly available/open research dataset used under its published terms; 70 sequences were used: 30 FALL and 40 ADL/NORMAL.<br><br>***In-house webcam dataset*** – 93 clips from 2 volunteers: 30 FALL and 63 NORMAL. Before recording, both volunteers were given a Participant Information Sheet and an Informed Consent | Field study at the Love and Elder Care Home: handwritten care records and conversations with caregivers (no audio recorded).<br><br>Svarah corpus (Indian-accented English speech): 6,656 utterances from 117 speakers, about 9.6 hours.<br><br>DisfluencySpeech and DISCO (English) corpora for disfluency removal.<br><br>25 domain-specific caregiving disfluency pairs (for example, medication self-corrections).<br><br>De-identified ADL dataset of 2,325 caregiver observation records in 8 | **WLASL**: public word-level ASL video dataset, used for research under its published terms.<br><br>**ASL Citizen**: public, community-sourced ASL dataset recorded on everyday webcams, used under its published terms.<br><br>59 care-related signs in 7 categories, chosen from caregiver needs discussed at CCHSS. This gave 2,352 source videos.<br><br>Industry input: visits to CCHSS (22 April 2026) and |

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| | | Ms. Judy Genovese, USC Stevens Center for Innovation). Files were downloaded from the provider's secure FTP server (FTP over TLS) and stored in Google Drive.<br>**EMA response file**: 14,795 survey rows from 212 participants. One daily survey with the stress question "Overall, how would you rate your current level of stress?" (1-5). Used as the outcome label (3 or above = elevated stress).<br>Fitbit heart-rate files (wrist photoplethysmography), one per participant, used for same-day and historical heart-rate features.<br>**Fitbit daily summary files**: steps, activity-zone minutes, resting heart rate and sleep efficiency.<br>Participant information file, used only for joins and splitting, not as a model input.<br>Participants were allocated before any feature was built (seed 42): 140 training, 30 validation, 27 test and 15 | Form, and consent was obtained before participation. The recordings were collected using a USB RGB webcam. | categories, generated with LLM support from the care-home information and manually checked.<br><br>200-case safety-focused test set, with expected labels reviewed by a caregiver. | Remark Elder Care (7 April 2026).<br><br>Parkinson's module: no patient data and no Parkinson's dataset |

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| | | demonstration, which was kept out of all modelling. 189 participants had usable rows (8,692 rows). | | | |
| | **Data Preprocessing:** | Survey timestamps were parsed, surveys with no stress rating were removed, and each valid survey's send time was used as the prediction cut-off.<br>Same-day features: from Fitbit heart-rate readings on the survey day taken strictly before the send time, the mean, standard deviation, minimum and maximum were computed. Surveys with no earlier reading were skipped.<br>Personal baseline: the mean of each person's daily heart rate over earlier days only. The last 3 and last 7 available daily deviations from this baseline were averaged (hr_dev_roll3, hr_dev_roll7).<br>Previous-day features from the Fitbit daily summary: steps, cardio, fat-burn, peak and out-of-range minutes, and resting heart rate. Sleep efficiency was taken from the survey date, as the previous night's sleep. | NTU Kinect skeletons and video-based skeletons were converted to a common 14-joint representation.<br><br>UR and in-house webcam videos were processed using MediaPipe Pose.<br><br>Every sample was converted to a 90-frame sequence (~3 seconds at 30 FPS) with 4 channels: x, y, z and visibility.<br><br>Hip-centred and torso-scale normalization was applied.<br>18 biomechanical features were extracted from each window.<br><br>Training data augmentation included temporal speed jitter, sub-window crop/resampling, left-right flip, yaw rotation, focal/roll jitter, z-channel drop with planar noise and Gaussian joint noise.<br><br>Features were recomputed after augmentation, so the ST-GCN and | Typed the handwritten care information manually, removed care recipients' names and ages, and assigned codes P01 - P41.<br><br>Checked all generated ADL records against the handwritten records and the caregivers' information, and removed duplicates.<br><br>Converted each ADL record into a 19-field target string with an alert label.<br><br>Resampled Svarah audio to 16 kHz and split it 80:10:10 (5,324 / 666 / 666) with seed 42.<br><br>Split DISCO (English) and the ADL dataset 80:10:10 with seed 42 (ADL: 1,860 / 232 / 233).<br><br>Repeated the 20 domain training pairs five times so they were not lost in the larger general data. | MediaPipe Holistic 0.10.14 extracted 49 landmarks per frame (both hands and upper body) × x, y, z = 147 values<br><br>Missing hand frames were repaired by forward fill, bringing missing frames from 23.7% down to 8.2%.<br>Landmarks were normalised relative to the body. Wrist-relative normalisation had dropped accuracy to about 5% on care signs.<br><br>Each sequence was resampled to 60 frames, and velocity and acceleration were added, giving 60 × 441 features.<br><br>3 videos failed extraction (2,349 left). An outlier rescue split long multi-repetition videos into single signs (+15), giving 2,364 final sequences.<br><br>Stratified 70/15/15 split by sign with random state 42 |

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| | | steps_deviation: previous-day steps minus the person's study-wide mean.<br>lag1_stress: the rating from the person's previous answered survey (1-5). Each person's first survey has no earlier answer and was removed (189 rows).<br>Stage 1 LightGBM regressor predicts the expected heart rate from hr_dev_roll3, hr_dev_roll7 and day of week; the difference between the observed and expected heart rate becomes hr_mean_deviation_model.<br>Missing values were filled with the training-set median, fitted on training people only and applied to validation and test.<br>Class imbalance was handled with a positive-class weight (negatives divided by positives) from the training labels only (elevated share 0.2398).<br>Participant identity and the current stress rating were excluded from the model. Final input: 16 features. | handcrafted-feature branches remained consistent. | Shuffled the speech training data before taking subsets, after finding a data-ordering problem. | (test set = 355). The scaler was fitted on the training set only.<br><br>Training data was augmented with three copies of each sequence, using random rotation (±8°), scaling (±5%) and temporal speed jitter (±15%), with seed 42. |

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| | **Data storage and versioning** | Raw TILES-2018 files, the cached tables and the saved participant allocation (seed 42, saved as a parquet file and reloaded, never regenerated) are stored in Google Drive.<br>The verified EMA table is saved as a parquet file, so every notebook starts from the same checked copy.<br>Research outputs are written to a separate research folder, so research runs cannot overwrite the files the application reads.<br>Saved outputs: the Stage 1 model, Stage 2 classifier and imputer (three joblib files), the SHAP plot, the repeated-validation results, the matched-model comparisons and the sensitivity-analysis files.<br>Application data is stored in Supabase PostgreSQL. Principal tables: participants (anonymized metadata and split group), daily_features (one row per participant and date, with pipeline version and score columns), participant_baselines, | Processed NTU, UR and webcam samples were saved as NumPy arrays, together with labels, subject IDs and split-index files.<br><br>Original volunteer .mp4 files were retained locally only, were not uploaded to cloud storage, and were not shared publicly.<br>Model checkpoints, scalers, threshold files and JSON evaluation reports were stored as project artefacts.<br><br>Training/evaluation scripts and processed artefacts were maintained in the project repository using Git/version control.<br><br>Runtime event and alert records are stored in Supabase PostgreSQL.<br><br>Optional privacy-preserving skeleton replay JSON for HIGH alerts is stored in Cloudflare R2 | Datasets, trained models and experiment results are stored in a dedicated project folder in Google Drive.<br><br>Git is used for version control of my component's source code, to track changes and keep a history of updates throughout development.<br><br>A fixed random seed (42) is used for all data splits, shuffling and sampling, so every data split can be reproduced.<br><br>The results of every experiment are saved as CSV files, so they can be compared and verified later.<br><br>The final trained models are saved as separate artefacts: final_whisper_lora_adapter (Whisper-small LoRA adapter), final_bart_disfluency (BART-base disfluency model) and STAGE5_RICH_FINAL_MODEL (T5-small ADL extraction model).<br><br>In the deployed system, structured ADL records and alerts are stored in Supabase (PostgreSQL) and | 3 Google Colab notebooks (01 data, 02 training, 03 DTW/ feedback/ learner model). Every result traces to a CSV or JSON file in one Google Drive evidence folder.<br><br>Original outputs are kept read-only, and reruns go into separate folders for comparison. Fixed seed 42.<br><br>The product bundle is versioned with a fingerprint per file (manifest.json). It also holds the Phase-1 scaler, label encoder, v3 thresholds, medoid references and parity files(pipeline_parity.npz).<br><br>Code is versioned with Git in the shared CareSense repository. The GLOSS database tables come from numbered Supabase migrations 0019–0024.<br><br>Medoid reference sequences are stored as JSONB in Supabase PostgreSQL. |

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| | | caregiver_profiles, model_registry (model name, version, training time, metrics, artifact paths) and audit_log.<br>Schema changes are kept as ordered migrations (43 migration files). Row-level security protects sensitive tables.<br>Model versions are recorded in the model_registry, so the app can state which model produced each score.<br>Library versions recorded: pandas 2.2.3, NumPy 2.1.3, scikit-learn 1.6.1, LightGBM 4.6.0, XGBoost 3.4.1, SHAP 0.52.0.<br>Code (notebooks and application source) is versioned in a GitHub repository: R26-DS-010-Intelligent-Care-Support. | | audio clips are stored in Cloudflare R2 object storage. | Reference sign videos and Parkinson's videos are in Cloudflare R2.<br><br>Learner videos are not stored, only the attempt results. |
| **Model Development** | **Model Selection** | Ablation 1 (test ROC AUC): same-day features only 0.5081, then pre-survey features 0.5361, adding the previous stress rating 0.6526, regularisation and early stopping 0.7086, learned expected heart rate (Stage 1) 0.7211. | Main model: Late-fusion classifier combining a 128-dimensional ST-GCN embedding with 18 biomechanical features.<br><br>ST-GCN was selected because it models spatial body-joint relationships and temporal movement patterns. | Speech recognition: Whisper-small with LoRA. Compared ranks 4, 8, 16 and 32 and selected rank 16 (WER 15.35%, close to rank 32 with half the trainable parameters).<br><br>Disfluency removal: compared a rule-based baseline, zero-shot models, and fine-tuned T5-small | Early stacked GRU, BiLSTM and CNN+BiLSTM models (774 features) overfitted. The best reached 67.18% on the earlier 66 signs.<br><br>Under a common parameter budget, three models were compared: BiLSTM- |

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| | | Ablation 2, previous stress rating: removing lag1_stress dropped the test AUC from 0.7213 to 0.6108. On rows with a previous rating, the full model scored 0.7273 and the previous rating alone 0.7026 (difference +0.0247, 95% CI -0.0010 to +0.0515).<br>Ablation 3, proximity centrality: adding degree and weighted centrality gave no detectable gain in random forest, XGBoost or LightGBM (all differences small, all intervals include zero).<br>Ablation 4, steps deviation: rebuilding it from earlier days only gave test AUC 0.7213 (CI 0.6650 to 0.7692), nearly the same as the main run.<br>Alternatives tested without the previous rating (LightGBM, XGBoost, CatBoost, logistic regression, SVM, neural network, extra trees, extra heart-rate window features): the best validation AUC was 0.5623, so none replaced the final model.<br>**Selected**: A two-stage design with one shared model for all | Handcrafted biomechanical features were retained because they provide explicit measures such as hip motion, torso angle, bounding-box shape and micro-movement instability.<br><br>Late fusion allowed the learned and engineered representations to be evaluated independently and then combined.<br><br>Baselines/comparators: ST-GCN-only, feature-only neural network, and Random Forest with 300 trees. | and BART-base. Selected BART-base because it resolved more safety-relevant self-corrections, such as medication doses and times.<br><br>ADL extraction: compared T5-small and BART-base. Selected T5-small (92.3% exact match, 100% alert recall on the test split).<br><br>Added field-based alert rules after T5-small, because most missed alerts conflicted with the model's own extracted fields. | Attention, TCN and a SPOTER-style transformer. TCN and SPOTER-style were then improved with 441 kinematic features.<br><br>Final model: the average of the softmax outputs of the improved TCN and improved SPOTER-style, chosen by validation macro-F1 only (0.8116), with 202,102 parameters.<br><br>Test results: macro-F1 0.7752 (95% CI 0.712–0.802) and accuracy 0.7859. That is 0.038 above the best single model (paired bootstrap CI 0.008–0.072, better in 99.5% of resamples).<br><br>Execution quality: DTW (dtw-python, symmetric2) against a medoid reference, chosen because it tolerates signing speed.<br><br>Feedback: rule-based, with Gemini allowed only to reword it under a guard. |

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| | | caregivers. Stage 1 is a LightGBM regressor that predicts each person's expected heart rate. Stage 2 is a LightGBM classifier that predicts elevated stress (rating 3 or above) from 16 features. Gradient-boosted trees were chosen because they work well on tabular data, are fast and work with SHAP explanations. | | | |
| | **Model Training** | Trained in Google Colab (Python notebooks) on the 140 training caregivers only. The 30 validation caregivers decided when training stopped, and the 27 test caregivers were used only for final assessment.<br>Stage 1: 100 trees, max depth 3, learning rate 0.05, seed 42, fitted on 5,977 training rows.<br>Stage 2: max depth 3, learning rate 0.05, min child samples 30, L1 and L2 penalty 0.5, subsample and column sampling 0.8, seed 42, up to 500 rounds.<br>Early stopping after 20 rounds on validation AUC (never test | Neural models implemented in PyTorch.<br><br>Main training used AdamW, learning rate $3 \times 10^{-4}$, weight decay $1 \times 10^{-4}$, batch size 32, cosine annealing, maximum 60 epochs and early stopping with patience 10.<br>Fixed random seed: 42.<br>Training loss = fusion cross-entropy + 0.2 × ST-GCN auxiliary cross-entropy.<br><br>Class imbalance handled using class-weighted cross-entropy and WeightedRandomSampler.<br><br>UR samples received 10× sampling weight because UR had | Whisper-small + LoRA on the q_proj and v_proj layers: alpha = 4 × rank, dropout 0.05, 3 epochs, batch size 8, mixed precision (fp16). WER fell from 26.13% to 15.57%.<br><br>BART-base for disfluency removal: learning rate 2e-5, weight decay 0.01, 3 epochs, batch size 8.<br><br>T5-small for ADL extraction: prefix “extract:”, learning rate 3e-4, 5 epochs, batch size 8, maximum length 192.<br><br>All training and evaluation done in Google Colab notebooks. | TensorFlow/Keras in Google Colab (GPU).<br><br>Adam optimiser, learning rate $1 \times 10^{-3}$, batch size 32, maximum 100 epochs, early stopping on validation macro-F1 with patience 12 and best weights restored.<br><br>Loss: focal loss ($\gamma = 2$) with label smoothing 0.1 and class-balanced weights;L2 regularisation $1 \times 10^{-4}$ and dropout.<br><br>TCN: three dilated causal convolution blocks (48 filters, kernel 3, dilations 1/2/4). SPOTER-style: two |

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| | | AUC); training stopped at round 11.<br>Class imbalance handled with scale_pos_weight (negatives divided by positives, from training labels only; elevated share 0.2398). The median imputer was also fitted on training data only.<br>Overfitting check: the first unregularized model scored 0.8654 on training and 0.6526 on test. Shallower trees and penalties cut the train-test gap to 0.0117. Final AUC: train 0.7328, validation 0.6874, test 0.7211.<br>Uncertainty: 95% confidence interval from a caregiver-level bootstrap (0.6644 to 0.7699), and 25 repeated participant-grouped validation fits (mean 0.6991, SD 0.0294).<br>Final models saved as three joblib files (Stage 1, Stage 2, imputer) and recorded in the model registry. | only 70 sequences compared with 3,777 NTU sequences.<br><br>Webcam adaptation used Subject 1's 35 clips for fine-tuning at learning rate $1 \times 10^{-5}$; Subject 2's 58 clips were used for evaluation/selection.<br><br>Thresholds were selected from validation data using a sweep from 0.05–0.95 based on macro-F1. | Evaluation measures: WER and CER; BLEU, ROUGE-L and exact match; field-level accuracy; alert recall and precision; a separate 200-case safety evaluation | transformer encoder layers (64-dimensional, 4 heads) with a class token. Stage A baselines on 147 features; Stage B on 441 kinematic features with augmentation; Stage C softmax averaging.<br><br>The test set was used once per model, with 95% CIs from 2,000 bootstrap resamples. Retraining from scratch gave 0.778 (original 0.775).<br><br>DTW thresholds were recalibrated with the live distance function (v1 → v3, n = 708). |
| **Model Integration** | **Tools used** | Python 3.11 – research notebooks, training, evaluation and backend. | Python 3.11 – backend, training and evaluation.<br><br>PyTorch – ST-GCN, fusion and feature-only neural networks. | Python 3.11 as the main programming language.<br><br>PyTorch as the deep learning framework. | Python 3.11 for training, evaluation and back end.<br><br>TensorFlow / Keras – TCN, |

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| | | pandas 2.2.3 / NumPy 2.1.3 – feature building, survey-cutoff windows, baselines and lag features.<br><br>scikit-learn 1.6.1 – median imputer, class-weight setup and evaluation metrics (ROC AUC, accuracy).<br><br>LightGBM 4.6.0 – Stage 1 expected-heart-rate regressor and Stage 2 elevated-stress classifier.<br><br>XGBoost 3.4.1 / random forest – matched comparison and ablation experiments.<br><br>SHAP (TreeExplainer) – factors that raise or lower each stress estimate (0.52.0 in research, 0.51.0 in the backend).<br><br>SciPy 1.16.3 / statsmodels 0.15.0 – statistical tests and caregiver-level bootstrap intervals. | MediaPipe Pose – extraction of 33 body landmarks from RGB video.<br><br>OpenCV – webcam capture, DirectShow integration, face blur, MJPEG encoding and overlays.<br><br>NumPy / scikit-learn – preprocessing, StandardScaler, Random Forest, sampling and evaluation metrics.<br><br>FastAPI + Uvicorn – REST API, WebSockets and video streaming.<br><br>Supabase PostgreSQL – rooms, patients, fall events and fall alerts.<br><br>Cloudflare R2 – optional skeleton replay storage.<br><br>React 19 + TypeScript + Vite + Tailwind CSS + Zustand + Recharts – SENTRY web dashboard.<br><br>pytest – automated backend testing. | Hugging Face Transformers, Datasets and PEFT (LoRA) for loading, fine-tuning and running the Whisper-small, BART-base and T5-small models.<br><br>JiWER for calculating word error rate (WER) and character error rate (CER).<br>LanguageTool for grammar correction, with a protected vocabulary of medication and food names.<br><br>gTTS for creating the spoken handover summary.<br><br>FastAPI for the backend that runs the models, and React + Vite for the web application.<br><br>Supabase (PostgreSQL) for storing ADL records and alerts, and Cloudflare R2 for storing audio clips.<br><br>Google Colab for model training and testing, and Google Drive for storing datasets, models and results. | SPOTER-style and the single-file ensemble.<br><br>MediaPipe Holistic 0.10.14 – hand and body landmarks (runs as a subprocess in its own environment).<br><br>NumPy / pandas / scikit-learn – preprocessing, metrics and bootstrap.<br><br>dtw-python – execution scoring.<br><br>Google Gemini API (gemini-3.5-flash-lite)– optional feedback rewording.<br><br>FastAPI + Uvicorn – REST API.<br>Supabase PostgreSQL and Auth – attempts, mastery, quiz results, login.<br><br>Cloudflare R2 – reference and Parkinson's videos.<br><br>React + Vite + TypeScript + Tailwind CSS – Sign & Vitals screens. |

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| | | joblib 1.5.3 – saving and loading the Stage 1 model, imputer and Stage 2 model.<br><br>FastAPI 0.141.1 + Uvicorn 0.52.4 – REST API for scoring, history, analytics, upload and redistribution.<br><br>httpx 0.28.1 / python-dotenv 1.2.3 / python-multipart 0.0.20 – HTTP calls, environment settings and file upload.<br><br>Supabase 2.31.0 (PostgreSQL and Auth) – daily features, scores, model registry, recommendations, roles and login, with row-level security.<br><br>React + TypeScript + CSS modules – PULSE views (Overview, Risk Analysis, Caregiver Profiles, Trends, Redistribution, Demo Control) inside the CareSense app.<br><br>Google Colab and Google Drive – model training and | | | Git/GitHub, Google Colab and Google Drive |

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| | | testing, and storage of TILES data, models and results.<br><br>Git/GitHub – version control for notebooks and application source. | | | |
| **Model Deployment** | **Testing Environments** | The model was tested on 27 held-out caregivers (1,406 survey responses) who were never used for training or tuning. The test ROC AUC was 0.7211, with a 95% confidence interval of 0.6644 to 0.7699 from resampling caregivers.<br><br>The model was also fitted 25 times on different splits of the development caregivers, giving an average AUC of 0.6991.<br>Checks confirmed that no caregiver appears in more than one group, the 15 demonstration caregivers were excluded, and the imputer and models were fitted on training caregivers only.<br>Ablation tests re-ran the model without the previous stress rating, with a historical-only | Random-split evaluation on the combined NTU+UR corpus.<br><br>Subject-disjoint evaluation on 6 unseen NTU subjects as the main benchmark.<br><br>Cross-dataset test: NTU-trained models tested directly on UR without adaptation.<br><br>Webcam adaptation test: fine-tuning on Subject 1 and evaluation on Subject 2.<br>Simulated ward live testing using a Hikvision DS-U02 USB RGB camera.<br><br>Manual developmental testing included sitting, standing, transitions and staged falls.<br><br>49 automated pytest tests covered feature extraction, post-processing, model behaviour and REST endpoints. | Google Colab T4 GPU: a 51.0-second recording processed in 12.2 seconds on average (3 runs).<br>Google Colab CPU runtime (single thread): the same recording processed in 77.2 seconds on average (2 runs), with the same records and alerts as on the GPU.<br><br>End-to-end tests on recordings spoken by the author, plus 100 multi-observation test paragraphs.<br><br>Verification test cases for data splits, output parsing, the protected vocabulary, long recordings and report consistency. | Held-out test set (355 videos) in Colab, with bootstrap confidence intervals and confidence-band analysis.<br><br>Latency over 5 Colab runs: single-file ensemble 52–97 ms; full attempt 73–144 ms.<br><br>Feedback faithfulness audit on 59 cases (guard v1 and v2). App parity tests: model 10/10 (max difference 2.4e-07), full pipeline 10/10, learner model 29/29 events.<br><br>23 test cases covering API security, low confidence, wrong sign, LLM failure, missing camera and invalid sign ID.<br><br>Live check with 6 real Gemini calls. |

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| | | steps feature, and with proximity centrality added.<br>A batch script scored every historical date through the live scoring route (1,487 rows), proving the web app, backend and model work end to end.<br>20 application test cases (TC1-TC20) covered roles and access, scoring, missing data, risk history, redistribution rules, uploads and start-up failure when a model file is missing.<br>Tested locally first (backend on port 8000, web app on port 5173), then on the hosted Render and Vercel deployment. | CPU latency benchmark performed over 100 runs on an AMD Ryzen 7 CPU. | | Live webcam end-to-end tests on a laptop. |
| | **Deployment Platform** | Deployed as one module of the CareSense web application, opened in the browser through the shared login.<br><br>The FastAPI backend, which loads the models and produces scores, is hosted on Render.<br>The React and TypeScript front end is hosted on Vercel. | Deployed on a Windows consumer laptop with a USB RGB camera.<br><br>Inference runs locally on the CPU, with no GPU required for the measured classification stage.<br><br>Backend runs using FastAPI/Uvicorn.<br><br>SENTRY web interface runs in React/TypeScript. | My component is deployed as a web application built with React + Vite, with a FastAPI backend that runs the three trained models (Whisper-small with LoRA, BART-base and T5-small)<br><br>It is integrated with the shared CareSense data layer, using Supabase (PostgreSQL) for ADL records and alerts and Cloudflare R2 for audio clips | Runs as the “Sign & Vitals” module inside the CareSense React web app, with a FastAPI back end and the shared Supabase project.<br><br>Reference videos are served from Cloudflare R2.<br><br>Currently runs locally on a laptop (Vite + Uvicorn). |

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| | | The database and login run on Supabase (PostgreSQL, Supabase Auth and row-level security).<br><br>Secrets such as the Supabase service-role key are set as environment variables on the host, not in the source code. | SENTRY integrates into the wider CareSense platform through the shared backend/database environment. | | Planned pilot hosting: a container with at least 2 GB RAM (1 CPU / 2 GB server); no GPU needed. |
| | **Deployment Method** | The code is kept on GitHub. The backend is deployed on Render and the front end on Vercel, both connected to the same Supabase project.<br><br>The three final model files (Stage 1 model, imputer, Stage 2 classifier) are placed in the backend's models folder and loaded once at start-up. The service will not start if one is missing.<br><br>A user signs in through Supabase Auth, and every request carries a token that the backend checks, together with the user's role.<br><br>To score a caregiver, the backend builds the day's features, runs the models and | Camera frames were captured locally.<br><br>MediaPipe extracts pose landmarks.<br><br>Landmarks mapped to the 14-joint skeleton.<br><br>A rolling 90-frame buffer is maintained.<br><br>Every third frame triggers inference once the buffer is full.<br><br>The sequence is normalized and passed through the ST-GCN and biomechanical-feature branches.<br><br>Late fusion produces the fall probability. | The caregiver records a voice log in the web application, and the application sends it to the FastAPI backend.<br><br>The backend splits the audio into pieces of up to 30 seconds and transcribes each piece with Whisper-small adapted with LoRA.<br>The transcript is split into separate sentences. BART-base removes fillers and spoken self-corrections, and LanguageTool corrects the grammar without changing medication or food names.<br><br>T5-small extracts a structured 19-field ADL record from each sentence, and the field-based alert rules decide whether the observation needs the supervisor’s attention. | The caregiver records a short webcam clip in the browser, and it is sent to POST /gloss/attempts with the Supabase token.<br><br>The server derives the caregiver profile from the token.<br><br>MediaPipe extracts landmarks, and the preprocessing is identical to the notebook.<br><br>GLOSS_ensemble.keras, loaded once at start-up, predicts the sign and its confidence.<br>Below 0.35 the learner is asked to repeat. Otherwise, if the sign matches the target, DTW grades the attempt and |

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| | | returns the stress probability, the predicted class and the SHAP factors shown on the PULSE pages.<br><br>Each model is recorded in a model registry table with its version, metrics and file path. | A temporal post-processing/state machine converts the score into NORMAL, MODERATE or HIGH.<br><br>Results are streamed to the dashboard through WebSockets and exposed through REST endpoints.<br><br>Events/alerts are written to Supabase.<br><br>HIGH alerts can store a privacy-preserving skeleton replay in R2. | The system creates a written report and a spoken handover summary using gTTS.<br><br>The extracted records and alerts are saved in Supabase and the audio clips in Cloudflare R2, and the supervisor dashboard reads the records and alerts from there. | locates the side and body part to correct.<br><br>Rule-based feedback is reworded by Gemini only if the guard accepts the reply.<br><br>Mastery and the next lesson are updated and saved in Supabase.<br><br>Parity checks run before each release. |
| **Future Enhancements** | **Model Improvement:** | Replace the single random split with several different splits, and keep a separate test group that is not looked at during development, so the final result is a fairer estimate.<br><br>Test the model on staff from care homes (with ethics approval and data-protection safeguards), because it has so far been trained and tested only on hospital workers.<br><br>Add new data sources, such as the OMsignal smart-shirt heart-rate variability and sleep | Collect a larger and more diverse in-house dataset, including more subjects, camera positions, clothing conditions and realistic activities.<br><br>Keep separate training, validation and untouched test subjects from the beginning.<br><br>Run multi-seed experiments and report model variance.<br><br>Improve domain adaptation between Kinect-based training data and MediaPipe RGB skeletons. | Train T5-small on inputs that contain several observations in one sentence, so that every event is extracted separately.<br><br>Split audio at natural pauses instead of fixed positions, so that sentences are not cut in the middle.<br><br>Add the care home's medication list to the speech recognition stage, so that drug names such as Losartan are recognised correctly.<br><br>Add numeric thresholds for vital signs, such as blood pressure and | Build a signer-independent test set that includes Sri Lankan caregivers.<br><br>Tune the confidence cut-offs on validation data, and recalibrate the DTW thresholds with novice attempts.<br><br>Test guard v2 on new Gemini replies.<br><br>Collect more data for weak signs such as hurt and pain.<br><br>Localise to Sri Lankan Sign |

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| | | records, after checking how complete they are for each caregiver.<br><br>Revisit the proximity (colleague co-location) features with better network coverage and a different network design, and test again whether they add anything. | Evaluate additional model architectures and run ablation studies for the ST-GCN branch, biomechanical features and post-processing rules.<br><br>Calibrate and validate alert thresholds using longer continuous recordings.<br><br>Measure true end-to-end latency, including pose extraction, database operations and network delivery.<br><br>Add multi-room and load testing.<br><br>Add authentication, restricted CORS, encrypted transport and stronger access control before real deployment.<br>Add input-drift monitoring to detect performance changes after camera/environment changes.<br><br>Conduct formal caregiver/clinical validation before any real ward use. | blood sugar values, and validate the alert rules on a new test set.<br><br>Reduce CPU processing time using model quantisation or faster inference methods.<br>Extend the system to support Sinhala, Tamil and code-mixed speech.<br><br>Add a wearable microphone so that caregivers can record observations hands-free while giving care. | Language with Sinhala and Tamil instructions.<br><br>Add a rigged 3D avatar and continuous real-time feedback.<br><br>Move MediaPipe into the browser and add a worker pool for multiple users.<br><br>Run the caregiver SUS evaluation.<br>Extend the symptom module to dementia and stroke. |